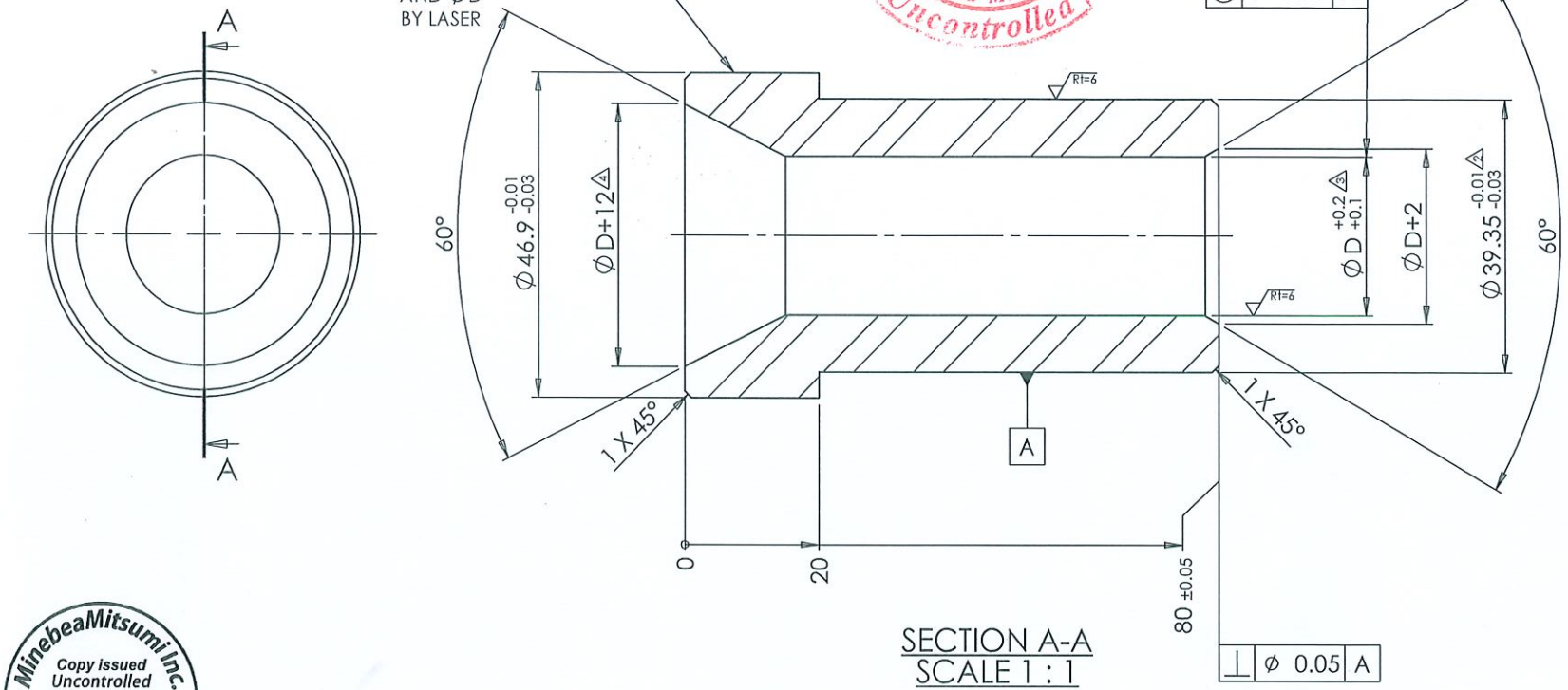
PART No. TGM-19-029A-1-xx | 4 REV.

Rt=12 (Rt=6)

Minebea Copy 27 JAN 2025 issued G/M and M/A Div. Uncontrolled

MARKING DWG. NO. #4
AND ØD
BY LASER



SECTION A-A
SCALE 1 : 1

ITEM	ØD	CHECK
-01	Ø23	Preecha
-02	Ø32	Preecha
-03	Ø21	Preecha
-04	Ø22	Preecha RQ :507/19
-05	Ø20	Preecha RQ :509/19
-06	Ø17	Preecha RQ :511/19
-07	Ø17.5	Preecha RQ :511/19
-08	Ø15.875	Preecha RQ : 91/20
-09	Ø18.3	Preecha RQ : 362/20
-10	Ø32.1	Preecha RQ : 487/20
-11	Ø32.2	Preecha RQ : 11/21
-12	Ø13.2	Preecha RQ : 326/21
-13	Ø17.1	Preecha RQ : 326/21
-14	Ø23.8	Preecha RQ : 292/22
-15	Ø11.1	Preecha RQ : 319/22
-16	Ø12.7	Preecha RQ : 319/22
-17	Ø9.525	Preecha RQ : 319/22
-18	Ø27	Preecha RQ : 308/23
-19	Ø21.5	Preecha RQ : 342/24
-20	Ø25	Preecha RQ : 15/25
-21		

MinebeaMitsumi Inc. Copy issued Uncontrolled 13/1/2568 Engineering Design center G/M.1 & M/A. Div.

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	13 Jun' 19	Preecha	Change : 39.4 ==> 39.35
3	----	30 Jul 19	Preecha	Change : $^{+0.1}_{+0.04}$ ==> $^{+0.2}_{+0.1}$ RQ : 340/19
4	----	12 Jun' 19	Preecha	Change : Ø38 ==> ØD+12

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S ~ UNCUT

M/C. B0326 III | Q'ty 1
MATERIAL SUS 303
FINISH -
HEAT TREATMENT -

List Request No. 91/20
SCALE : | SHEET / | UNIT (mm.)
3 RD ANGLE

DETAILED	DESIGNED	CHECKED	APPROVED
Chanon	Chonknok	Preecha	Preecha

DATE 2 Mar' 20

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Bush sleeve 1

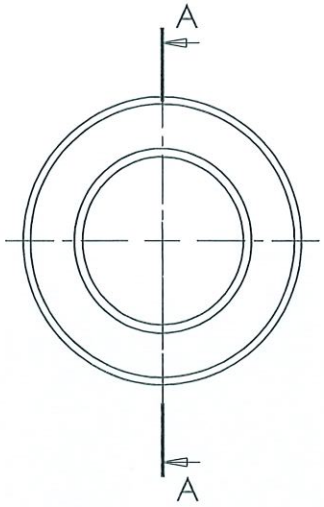
PART No. TGM-19-029A-1-xx | 4 REV.

PART No. TGM-19-029A-2-xx

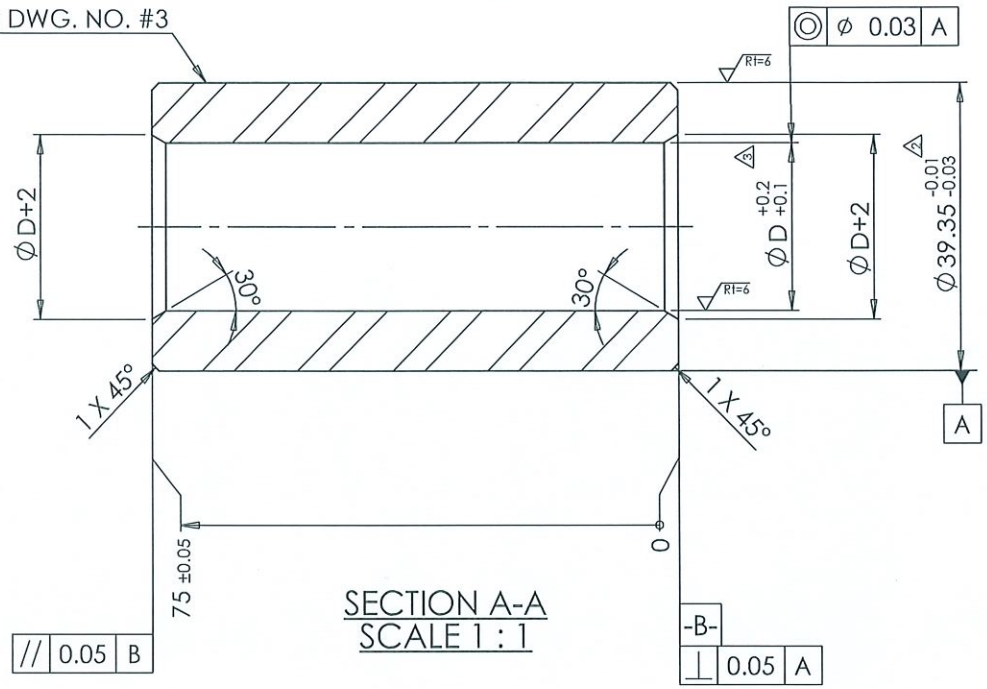
3 REV.

Rt=12 (Rt=6)

Minebea Copy 27 JAN 2025 Issued G/M and M/A Div. Uncontrolled



MARKING DWG. NO. #3
AND ØD
BY LASER



SECTION A-A
SCALE 1 : 1

ITEM	ØD	CHECK
-01	Ø23	Preecha
-02	Ø32	Preecha
-03	Ø21	Preecha
-04	Ø22	Preecha RQ :507/19
-05	Ø20	Preecha RQ :569/19
-06	Ø17	Preecha RQ :511/19
-07	Ø17.5	Preecha RQ :511/19
-08	Ø15.875	Preecha RQ : 91/20
-09	Ø18.3	Preecha RQ : 362/20
-10	Ø32.1	Preecha RQ : 487/20
-11	Ø32.2	Preecha RQ : 11/21
-12	Ø13.2	Preecha RQ : 326/21
-13	Ø17.1	Preecha RQ : 326/21
-14	Ø23.8	Preecha RQ : 326/21
-15	Ø11.1	Preecha RQ : 319/22
-16	Ø12.7	Preecha RQ : 319/22
-17	Ø9.525	Preecha RQ : 319/22
-18	Ø27	Preecha RQ : 308/23
-19	Ø21.5	Preecha RQ : 342/24
-20	Ø25	Preecha RQ : 15/25
-21		

MinebeaMitsumi Inc. Copy issued Uncontrolled 13/1/2568 Engineering Design center G/M.1 & M/A. Div.

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	13 Jun' 19	Preecha	Change : 39.4 ==> 39.35
3	----	30 Jul' 19	Preecha	Change : $^{+0.1}_{+0.04}$ ==> $^{+0.2}_{+0.1}$ RQ : 340/19

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S ~ UNCUT

M/C. B0326 III | Q'ty 5

MATERIAL SUS 303

FINISH -

HEAT TREATMENT -

List Request No. 91/20

3 RD ANGLE

SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Chanon	Chonknok	Preecha	Preecha

DATE 2 Mar' 20

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Bush sleeve 2

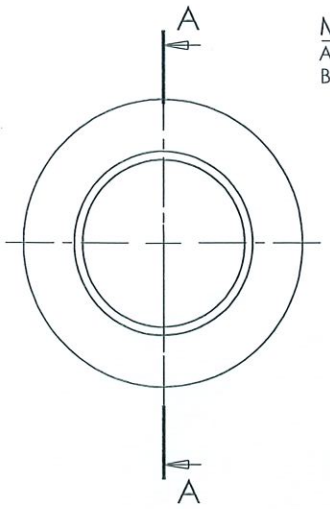
PART No. TGM-19-029A-2-xx

3 REV.

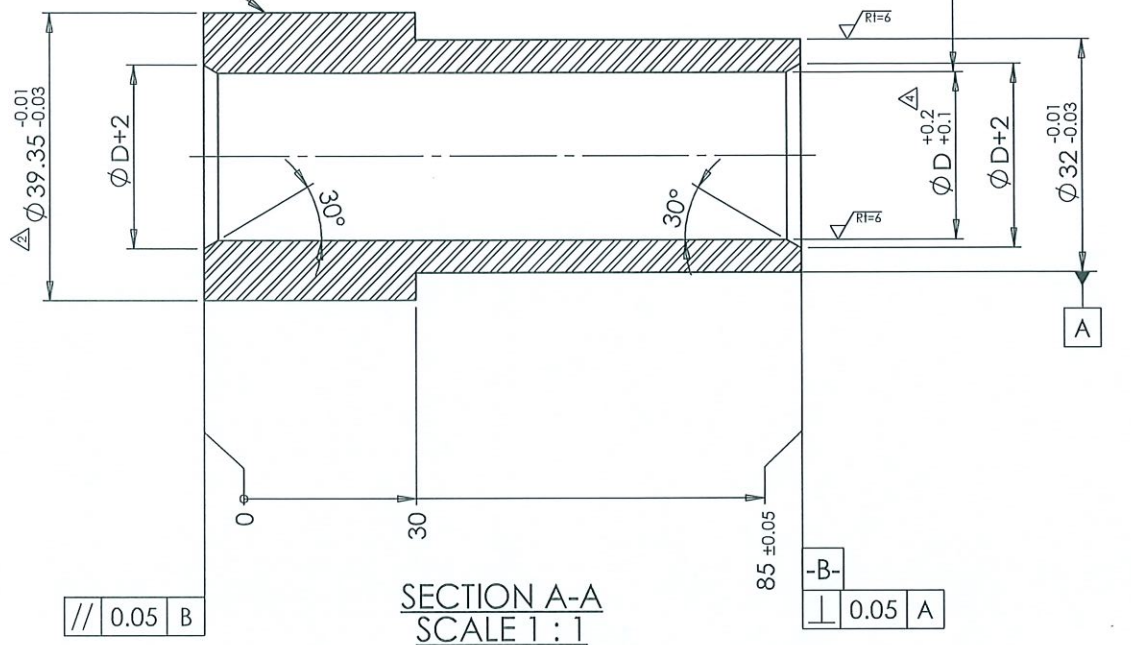
PART No. TGM-19-029A-3-xx

4 REV.

Rt=12 (Rt=6)



MARKING DWG. NO. #4
AND ØD
BY LASER



SECTION A-A
SCALE 1 : 1

Minebea Copy ... 2025 Issued G/M and M/A Div. Uncontrolled

MinebeaMitsumi Inc. Copy Issued Uncontrolled 13/1/2568 Engineering Design center G/M.1 & M/A. Div.

ITEM	ØD	CHECK
-01	Ø23	Freecha
-02	~~Ø32~~	Freecha
-03	Ø21	Freecha
-04	Ø22	Freecha RQ :507/19
-05	Ø20	Freecha RQ :509/19
-06	Ø17	Freecha RQ :511/19
-07	Ø17.5	Freecha RQ :511/19
-08	Ø15.875	Freecha RQ : 91/20
-09	Ø18.3	Freecha RQ : 362/20
-10	Ø13.2	Freecha RQ : 326/21
-11	Ø17.1	Freecha RQ : 326/21
-12	Ø23.8	Freecha RQ : 292/22
-13	Ø11.1	Freecha RQ : 319/22
-14	Ø12.7	Freecha RQ : 319/22
-15	Ø9.525	Freecha RQ : 319/22
-16	Ø27	Freecha RQ : 308/23
-17	Ø21.5	Freecha RQ : 342/24
-18	Ø25	Freecha RQ : 15/25
-19		

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	13 Jun' 19	Preecha	Change : 39.4 ==> 39.35
3	----	18 Jun' 19	Preecha	Delete : Ø32 Item 02
4	----	30 Jul' 19	Preecha	Change : $^{+0.1}_{+0.04}$ ==> $^{+0.2}_{+0.1}$ RQ : 340/19

UNLESS SPECIFIED C0.1 OR R0.1 GENERAL TOLERANCE JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S ~ UNCUT

M/C.	Q'ty
B0326 III	1

MATERIAL: SUS 303

FINISH: -

HEAT TREATMENT: -

List Request No. 91/20

3 RD ANGLE

SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Chanon	Chonknok	Preecha	Preecha

DATE 2 Mar' 20

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: BUSH SLEEVE 3

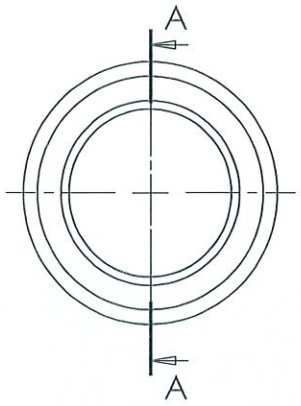
PART No. TGM-19-029A-3-xx

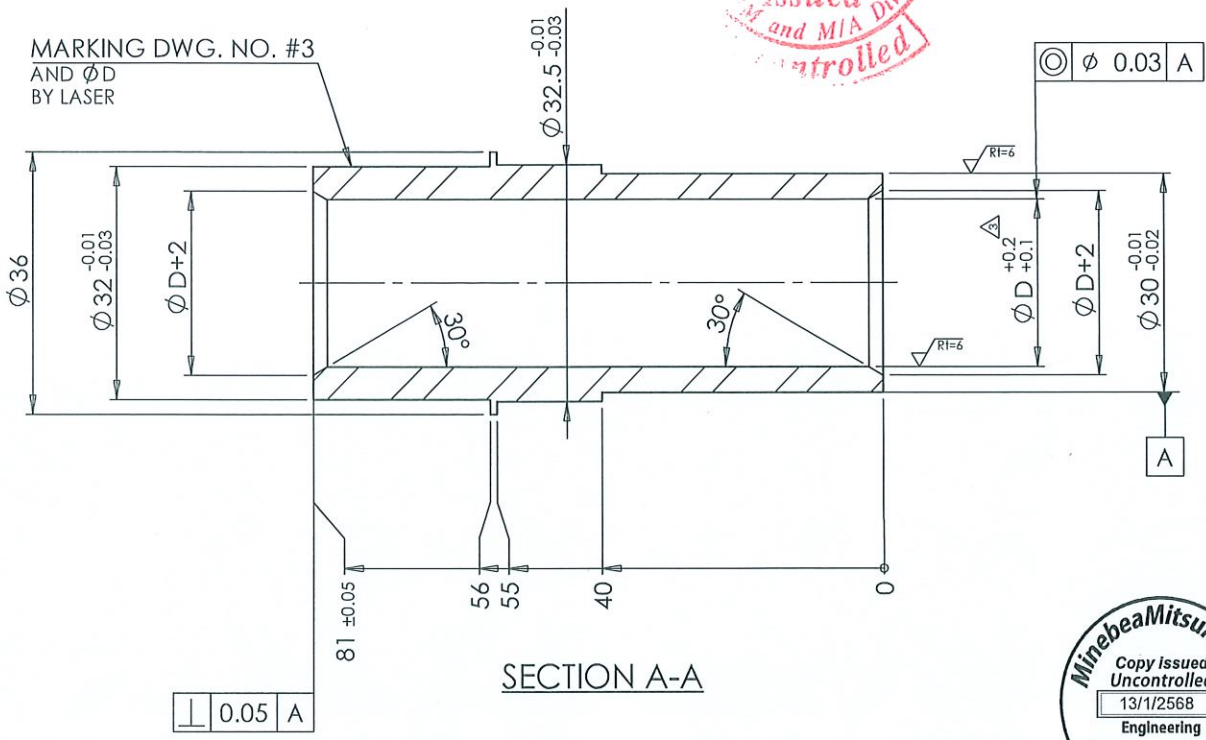
4 REV.

PART No. TGM-19-029A-4-xx

3 REV.

Rt=12 (Rt=6)





SECTION A-A

Minebea Copy 2025 Issued G/M and M/A Div. Uncontrolled

MinebeaMitsumi Inc. Copy issued Uncontrolled 13/1/2568 Engineering Design center G/M.1 & M/A. Div.

ITEM	ØD	CHECK
-01	Ø23	Preecha
-02	~~Ø32~~	
-03	Ø21	Preecha
-04	Ø22	Preecha RQ :507/19
-05	Ø20	Preecha RQ :509/19
-06	Ø17	Preecha RQ :511/19
-07	Ø17.5	Preecha RQ :511/19
-08	Ø15.875	Preecha RQ : 91/20
-09	Ø18.3	Preecha RQ : 362/20
-10	Ø13.2	Preecha RQ : 326/21
-11	Ø17.1	Preecha RQ : 326/21
-12	Ø23.8	Preecha RQ : 292/22
-13	Ø11.1	Preecha RQ : 319/22
-14	Ø12.7	Preecha RQ : 319/22
-15	Ø9.525	Preecha RQ : 319/22
-16	Ø27	Preecha RQ : 308/23
-17	Ø21.5	Preecha RQ : 342/24
-18	Ø25	Preecha RQ : 15/25
-19		

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	18 Jun' 19	Preecha	Change : 23 ==> ØD etc;.
3	----	30 Jul' 19	Preecha	Change : $^{+0.1}_{+0.04}$ ==> $^{+0.2}_{+0.1}$ RQ : 340/19

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S ~ UNCUT

M/C.	Q'ty
B0326 III	1

MATERIAL: SUS 303

FINISH: -

HEAT TREATMENT: -

List Request No. 91/20

3 RD ANGLE

SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Chanon	Chonknok	Preecha	Preecha

DATE 2 Mar' 20

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: BUSH SLEEVE 4

PART No. TGM-19-029A-4-xx

3 REV.

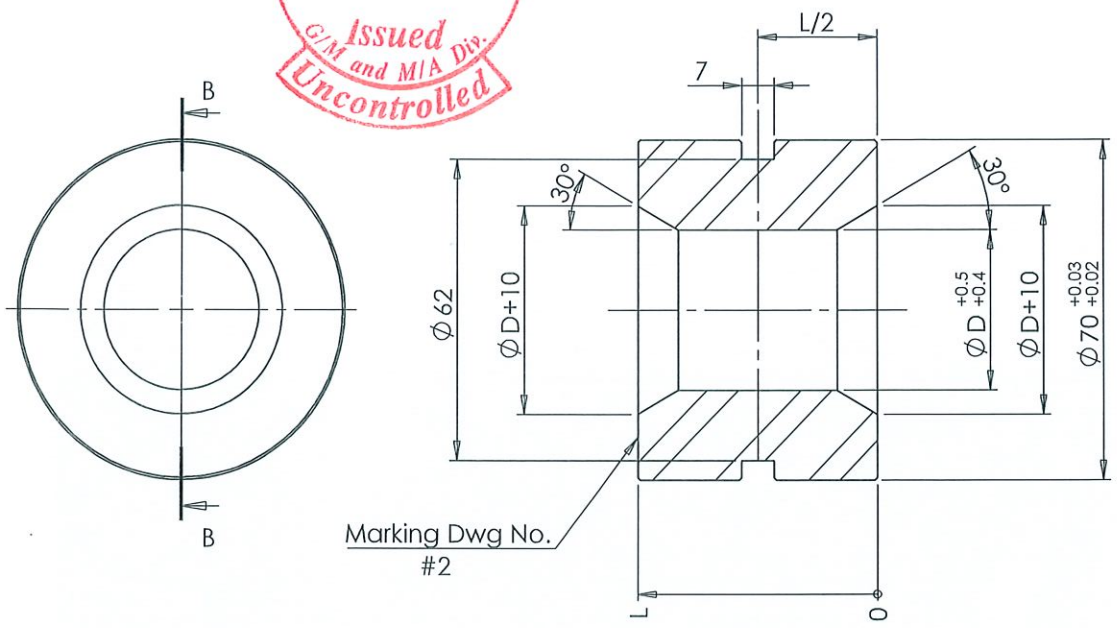
PART No. TGM-19-087-02-xx

2 REV.

Rt=12

Minebea Copy
27 JAN 2025
Issued G/M and M/A Div.
Uncontrolled

Formula : ØD = OD Mat"l +1



SECTION B-B
SCALE 1 : 1.5

Item	ØD	L	For Mat'l	Q'ty For:	Check
-01	Ø19	51.6	Ø18.3	5 Rotary	Preecha
-02	Ø19	91.6	Ø18.3	5 Guide Bush Less	Preecha
-03	Ø21	51.6	Ø20.5	5 Rotary	Preecha RQ:262/22
-04	Ø21	91.6	Ø20.5	5 Guide Bush Less	Preecha RQ:262/22
-05	Ø23	51.6	Ø22	5 Rotary	Preecha RQ:262/22
-06	Ø23	91.6	Ø22	5 Guide Bush Less	Preecha RQ:262/22
-07	Ø24	51.6	Ø23	5 Rotary	Preecha RQ:262/22
-08	Ø24	91.6	Ø23	5 Guide Bush Less	Preecha RQ:262/22
-09	Ø14	51.6	Ø13.2	5 Rotary	Preecha RQ:276/22
-10	Ø14	91.6	Ø13.2	5 Guide Bush Less	Preecha RQ:276/22
-11	Ø18	51.6	Ø17.1	5 Rotary	Preecha RQ:276/22
-12	Ø18	91.6	Ø17.1	5 Guide Bush Less	Preecha RQ:276/22
-13	Ø24.6	51.6	Ø23.8	5 Rotary	Preecha RQ:292/22
-14	Ø24.6	91.6	Ø23.8	5 Guide Bush Less	Preecha RQ:292/22
-15	Ø12.1	51.6	Ø11.1	5 Rotary	Preecha RQ:319/22

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

(Fedek SN-542-3.2)

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	5 JUL'22	Preecha	Ch : See Table => Pom black , etc. RQ:262/22

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JIS B0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S ~ UNCUT

M/C. - Q'ty -

MATERIAL Pom black ⚠

FINISH -

HEAT TREATMENT -

List Request No. 62/22

3 RD ANGLE

SCALE : SHEET / UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Chonknok	Chonknok	Preecha	Preecha

DATE 10 FEB'22

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Bush Sleeve (A) ⚠2

PART No. TGM-19-087-02-xx

2 REV.

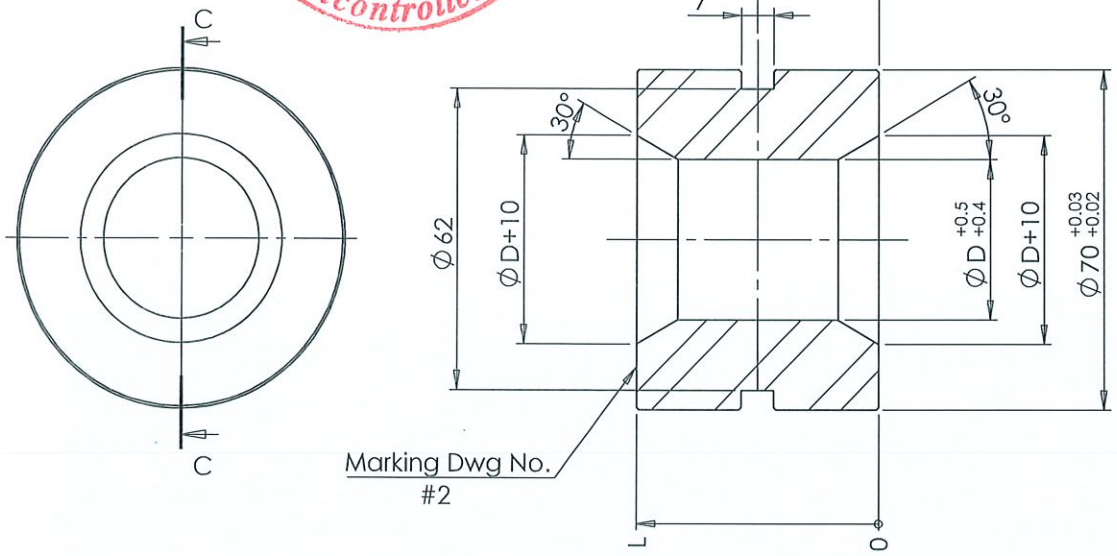
PART No. TGM-19-087-02-xx

2 REV.

Rt=12

Minebea Copy
27 JAN 2025
Issued G/M and M/A Div.
Uncontrolled

Formula : ØD = OD Mat"l +1



SECTION C-C
SCALE 1 : 1.5

Item	ØD	L	For Mat'l	Q'ty For:	Check
-16	Ø12.1	91.6	Ø11.1	5 Guide Bush Less	Preecha RQ:319/22
-17	Ø13.7	51.6	Ø12.7	5 Rotary	Preecha RQ:319/22
-18	Ø13.7	91.6	Ø12.7	5 Guide Bush Less	Preecha RQ:319/22
-19	Ø10.525	51.6	Ø9.525	5 Rotary	Preecha RQ:319/22
-20	Ø10.525	91.6	Ø9.525	5 Guide Bush Less	Preecha RQ:319/22
-21	Ø16.875	51.6	Ø15.875	5 Rotary	Preecha RQ:319/22
-22	Ø16.875	91.6	Ø15.875	5 Guide Bush Less	Preecha RQ:319/22
-23	Ø28	51.6	Ø27	5 Rotary	Preecha RQ:308/23
-24	Ø28	91.6	Ø27	5 Guide Bush Less	Preecha RQ:308/23
-25					
-26					
-27					
-28					
-29					
-30					

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

(Fedek SN-542-3.2)

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	5 JUL'22	Preecha	Ch : See Table => Pom black , etc. RQ:262/22

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S UNCUT

M/C. - Q'ty -

MATERIAL Pom black ⚠

FINISH -

HEAT TREATMENT -

List Request No. 62/22

3 RD ANGLE

SCALE : SHEET / UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Chonknok	Chonknok	Preecha	Preecha

DATE 10 FEB'22

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Bush Sleeve (A) ⚠2

PART No. TGM-19-087-02-xx

2 REV.